Data Mining HW2

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Q: How do you select features for your model input, and what preprocessing did you perform to review text?

- Feature selection:

  I select 'title' and 'text' as the input features for the pretrained BERT model, because the BERT model is specifically designed for processing textual data. These fields directly provide textual content, allowing the model to analyze and predict the sentiment of the texts.

- Data preprocessing:

  I. Filling Missing Values

  I filled missing values in the 'title' and 'text' columns with empty strings to avoid errors in subsequent data processing steps, as the model cannot handle null values.

  II. Combining Text

  I combined 'title' and 'text' into a single 'review' column to integrate the information from both parts, providing richer context to the model, which helps enhance the model's understanding of the overall text meaning.

  III. Using Tokenizer

  I utilized the BertTokenizer for tokenizing the text because it is specifically designed for the BERT model, transforming the text into a format that the model can understand and process.

  IV. Adjusting Text Length

  I set a maximum length of 512 for all texts, applying padding and truncation as needed. This was done to meet the model's input requirements, as the BERT model requires fixed-length inputs.

  V. Label Adjustment

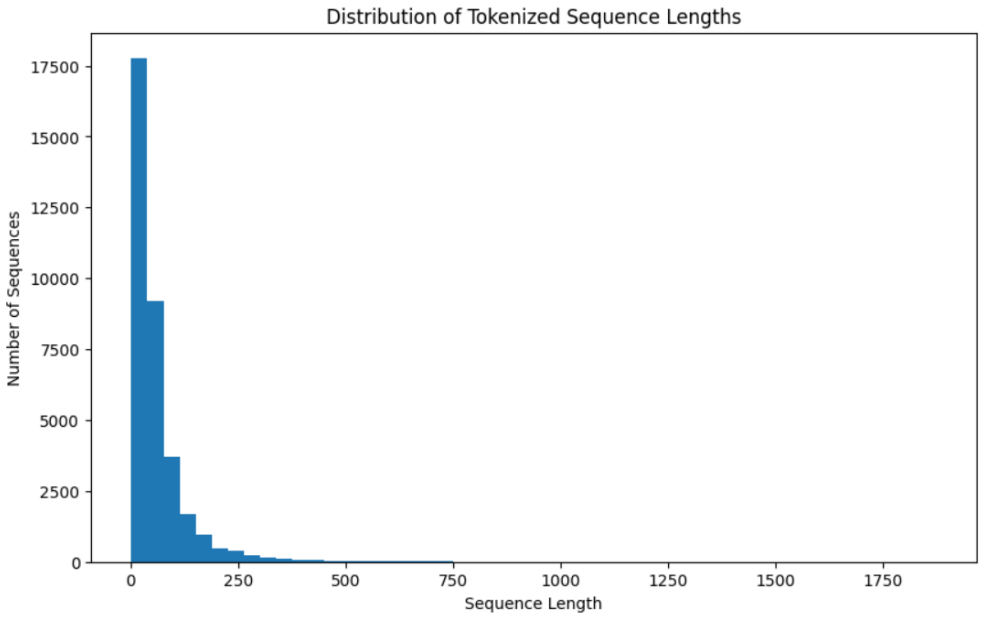
  I adjusted the 'rating' labels to start from 0 instead of 1 because the model's cross-entropy loss function requires zero-based label indexing.

Q: Please describe how you tokenize your data, calculate the distribution of tokenized sequence length of the dataset and explain how you determine the padding size.

- Tokenization of Data

  I used the BertTokenizer from the Hugging Face to tokenize reviews in the dataset. I start by combining the titles and texts into a single string, then use this tokenizer to transform them into token sequences. I'm using the bert-base-uncased model, which means it converts all English letters to lowercase. This tokenizer is trained on a model called WordPiece, which can identify the basic units of English, such as whole words or parts of words. This approach allows the model to better understand and handle different contexts and vocabularies.

- Calculation of Tokenized Sequence Length Distribution



To better understand the characteristics of the data and decide how to set the input length for the model, I calculated the original length of the token sequences from each review without truncating or padding them first. I then used this data to plot a histogram, which visually shows the distribution of token sequence lengths. This step allows us to see the typical lengths of the reviews, helping us to set the max_length for the model appropriately.

- Determination of Padding Size

  From the histogram and the data on sequence lengths, I noticed that the length of most reviews is much less than the maximum limit of the BERT model (512 tokens). Therefore, I chose 512 as max_length. This length not only covers the entirety of most reviews but also reduces memory consumption and computational time while ensuring training efficiency.

  I also tried using a max_length of 256 for tokenization, but it resulted in a public score of 0.6367, which is lower compared to the score of 0.6377 achieved by using a max_length of 512.

Q: Please compare the impact of using different methods to prepare data for different rating categories.

I tried three different data preparation methods to observe their impact on the model's ability to predict different rating categories. Specifically, I compared the following three configurations:

1. Title + Text Configuration F1 score: 0.6314
2. Title Only Configuration F1 score: 0.5330
3. Text Only Configuration F1 score: 0.5597

From these results, it's clear that combining the title and text as input features significantly enhances the model's capability to predict different rating categories. This is likely because the merged text provides more comprehensive information, helping the model to better understand and classify the reviews.